3 **(a)** Express $\dfrac{1}{(x+2)(x+3)}$ in partial fractions. **[3]**

(b) Find $\displaystyle\int \frac{1}{(x+2)(x+3)}\,\mathrm{d}x$ in the form $\ln\big(\mathrm{f}(x)\big)+c$, where c is the constant of integration and $\mathrm{f}(x)$ is a function to be determined. **[3]**